

Discrete mathematics

1. Language

Definition 1.1 (Semantics)

The meaning of a sequence of symbols.

Definition 1.2 (Syntax)

The grammatical rules for composing symbols into valid sentences.

Definition 1.3 (Variable)

A symbol that stands in place for an object that has not been determined yet.

A name can be assigned to a particular object using $:=$.

Definition 1.4 (Sentence)

A sentence is the expression of an idea in accordance of the syntax and grammar of a given language.

A statment is **atomic** if it cannot be broken down into smaller components while complying with the syntax and grammar.

- **Declarative:** Describes something. Usually, it describes a **subject** and a property, called **predicate**, that the subject has.
- **Interrogative:** Asks a non-rhetorical question.
- **Imperative:** Command or request.

In math, we are concerned with declarative sentences about mathematical objects.

1.1. Church-Turing thesis

Definition 1.1.1 (Computable)

A function is **computable** if any of the following equivalent things are true:

- It is expressible as a general recursive process.
- It is a term in the λ -calculus.
- It can be described by a Turing machine.

Axiom 1.1.2 (Church-Turing Thesis)

Our intuitive definition of “computable” (which Turing calls “effectively calculable”) is equivalent to the above definition.

2. Zeroth-order logic

2.1. Truth values

See Cedre DM1.1

Definition 2.1.1 (Truth value)

There are only two truth values:

- $\top$ “true”
- $\perp$ “false”

Definition 2.1.2 (Logical equivalence of propositions)

Two propositions p and q are logically equivalent, denoted $p \equiv q$, iff they have the same truth value.

Theorem 2.1.3

Logical equivalence is an equivalence relation.

Axiom 2.1.4 (Principle of Bivalence)

Sentences expressing truth values are either true or false, but not both.

Definition 2.1.5 (Proposition (informal definition))

A statement which is either true or false, i.e. a statement expressing a truth value.

2.2. Logical connectives

See Fleck 2.2~2.9

See Cedre DM1.2

Definition 2.2.1 (Negation)

$$\neg \top = \perp$$

$$\neg \perp = \top$$

Definition 2.2.2 (Logical dual)

Logical connectives f and g are logical duals iff $\neg f = g$ and $\neg g = f$ on all possible inputs.

Definition 2.2.3 (Conjunction, disjunction)

- **Conjunction:** $p \wedge q = p$ AND q
- **Disjunction:** $p \vee q = p$ OR q (inclusive or)

Conjunction and disjunction are dual.

Definition 2.2.4 (Conditional)

$p \rightarrow q$ means “if p , then q ”. p is the **antecedent** and q is the **consequent**.

If p and q are propositional formulae: $p \rightarrow q$ is true iff every time p is satisfied, then q is satisfied.

If p and q are propositions: it is false iff p is true and q is false.

Warning: If p is always false, then $p \rightarrow q$ is always true regardless if q is false! Also, there need not be any causal relationship between p and q .

Definition 2.2.5 (Converse)

The **converse** of $p \rightarrow q$ is $q \rightarrow p$. These are not equivalent.

Definition 2.2.6 (Biconditional)

$$p \leftrightarrow q = p \text{ iff } q$$

$$p \leftrightarrow q \implies (p \rightarrow q) \wedge (q \rightarrow p)$$

Definition 2.2.7 (Exclusive disjunction)

$$p \oplus q = p \text{ XOR } q = p \leftrightarrow .\neg q$$

p	q	$\neg p$	$p \wedge q$	$p \vee q$	$p \oplus q$	$p \rightarrow q$	$p \leftrightarrow q$
$\top$	$\top$	$\perp$	$\top$	$\top$	$\perp$	$\top$	$\top$
$\top$	$\perp$	$\perp$	$\perp$	$\top$	$\top$	$\perp$	$\perp$
$\perp$	$\top$	$\top$	$\perp$	$\top$	$\top$	$\top$	$\perp$
$\perp$	$\perp$	$\top$	$\perp$	$\perp$	$\perp$	$\top$	$\top$

Definition 2.2.8 (Proposition)

λ is a proposition iff λ satisfies the following recurrence:

- $\lambda = \top$ or $\lambda = \perp$.
- $\lambda = \neg\varphi$, where φ is a proposition.
- $\lambda = \varphi \wedge \psi$, where φ and ψ are propositions.
- $\lambda = \varphi \vee \psi$, where φ and ψ are propositions.
- $\lambda = \varphi \rightarrow \psi$, where φ and ψ are propositions.
- $\lambda = \varphi \leftrightarrow \psi$, where φ and ψ are propositions.

We use this definition inductively, by first establishing that $\top$ and $\perp$ are propositions, and then using the connectives to compose larger and larger propositions. Alternatively, we use this definition recursively to determine that statements are propositions by recursively decomposing them.

2.3. Propositional formulae

See Cedre DM1.2

Definition 2.3.1 (Propositional formula)

An expression that evaluates as a proposition when all of its variables are themselves replaced by propositions.

Definition 2.3.2 (Logical equivalence and nonequivalence)

Let φ and ψ be propositional formulae both consisting of the same variables $p_1, \dots, p_n$. $\varphi \equiv \psi$ iff every assignment of truth values to those variables produces the same truth value.

Theorem 2.3.3

Computing a truth table to check if two propositional formulae of n variables are equivalent requires 2^n rows.

Thus, we want to prove that the two expressions are equivalent, rather than brute-forcing them.

2.4. Propositional logic

Axiom 2.4.1 (Axioms of classical logic)

The following are the axioms of *Boolean algebra*, and they have two equivalent forms:

Identity	$\top \wedge p \equiv p$	$\perp \vee p \equiv p$
Complement	$\neg p \wedge p \equiv \perp$	$\neg p \vee p \equiv \top$
Commutativity	$p \wedge q \equiv q \wedge p$	$p \vee q \equiv q \vee p$
Associativity	$p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r$	$p \vee (q \vee r) \equiv (p \vee q) \vee r$
Distributivity	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	

2.5. More propositional logic

See Fleck 2.2~2.9

Theorem 2.5.1 (De Morgan's Laws)

- $\neg(p \wedge q) \equiv \neg p \vee \neg q$
- $\neg(p \vee q) \equiv \neg p \wedge \neg q$

Theorem 2.5.2 (Distributive laws)

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

Theorem 2.5.3 (Negation equivalences)

- $\neg(\neg p) \equiv p$
- $\neg(p \rightarrow q) \equiv p \wedge \neg q$

2.6. Predicates and variables